

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 17 — Carbonyl Compounds: Aldehydes & Ketones (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	17.1 Structure, Physical Properties & Nucleophilic Addition of HCN · 17.2 Diagnostic Testing: 2,4-DNPH, Tollens', Fehling's
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 17.1 Structure, Properties & Nucleophilic Addition** — Planar sp^2 hybridised carbonyl group ($>C=O$), strong permanent dipole-dipole attractions, water solubility via hydrogen bonding.
- 17.2 Diagnostic Testing & The Iodoform Reaction** — Condensation with 2,4-dinitrophenylhydrazine (Brady's reagent) forming yellow/orange/red crystalline precipitates.
- High-Frequency Core Repeats** — The 10 most frequently examined Cambridge Paper 1 questions on Carbonyl Chemistry (Q101 – Q110).

SUBTOPIC 17.1 — STRUCTURE, PROPERTIES & NUCLEOPHILIC ADDITION (Q1 – Q55)

1 [EASY] Nature of the Carbonyl Double Bond — 9701/11/M/J/23/Q26

[1]

Which statement correctly describes the bonding and charge distribution in the carbonyl group ($>\text{C}=\text{O}$)?

- [A] The carbonyl carbon is sp^3 hybridised with tetrahedral bond angles.
- [B] The carbonyl carbon is sp^2 hybridised and planar; the $\text{C}=\text{O}$ bond is strongly polarized ($\text{C}^{\delta+} = \text{O}^{\delta-}$) because oxygen is significantly more electronegative than carbon.
- [C] The $\text{C}=\text{O}$ bond is non-polar because electrons are shared equally in a π bond.
- [D] The carbon atom carries a partial negative charge ($\text{C}^{\delta-}$) which attracts electrophiles.

2 [HARD] Mechanism of Nucleophilic Addition of HCN to Carbonyls — 9701/12/M/J/23/Q26

[1]

In the addition of hydrogen cyanide, HCN, to ethanal, why is a catalytic amount of sodium cyanide (or sodium hydroxide) added to the reaction mixture?

- [A] Sodium cyanide acts as an oxidizing agent that oxidizes ethanal.
- [B] Sodium hydroxide neutralizes the toxic fumes of hydrogen cyanide.
- [C] Cyanide ions act as an electrophile that attacks the oxygen atom.
- [D] HCN is a very weak acid that provides a negligibly low concentration of cyanide ions ($:\text{CN}^-$); adding NaCN provides free $:\text{CN}^-$ nucleophiles to initiate attack on the carbonyl carbon.

3 [HARD] Stereochemical Consequence of HCN Addition to Unsymmetrical Carbonyls — 9701/13/M/J/23/Q26

[1]

When propanal ($\text{CH}_3\text{CH}_2\text{CHO}$) reacts with HCN in the presence of NaCN catalyst, what is the optical nature of the resulting 2-hydroxybutanenitrile product?

- [A] It is an optically inactive racemic mixture containing equimolar amounts of two enantiomers because the planar carbonyl group is attacked with equal probability from above or below.
- [B] It consists of a single pure optically active (R)-enantiomer.
- [C] It is optically inactive because the product molecule contains no chiral center.
- [D] It undergoes spontaneous inversion to form an achiral carboxylic acid.

4 [EASY] Reduction of Aldehydes and Ketones by NaBH_4 — 9701/11/O/N/23/Q26

[1]

What are the organic reduction products formed when butanal and butanone are separately treated with aqueous ethanolic sodium borohydride (NaBH_4)?

- [A] Both compounds reduce to butan-1-ol.
- [B] Both compounds reduce to butane.
- [C] Butanal reduces to butan-1-ol (primary alcohol); Butanone reduces to butan-2-ol (secondary alcohol).
- [D] Butanal reduces to butanoic acid; Butanone reduces to butan-2-ol.

5 [EASY] Boiling Point Comparison: Carbonyls vs Alkanes and Alcohols — 9701/13/M/J/22/Q5

[1]

Why does propanal ($M_r = 58.0$) boil at 49°C , which is higher than butane ($M_r = 58.0$, bp -0.5°C) but much lower than propan-1-ol ($M_r = 60.0$, bp 97°C)?

- [A] Propanal possesses permanent dipole-dipole attractions due to its polar $\text{C}=\text{O}$ bond, which are stronger than London forces in butane, but lacks intermolecular hydrogen bonds found in propan-1-ol.
- [B] Propanal decomposes at 50°C .
- [C] Butane forms intermolecular hydrogen bonds.
- [D] Propan-1-ol contains no polar bonds.

6 [HARD] Water Solubility of Lower Carbonyls — 9701/11/M/J/221/Q6**[1]**

Why are propanal and propanone completely miscible with water, whereas hexan-2-one is insoluble?

- [A] Propanone reacts with water to form propane.
- [B] Hexan-2-one is an ionic solid.
- [C] Water only dissolves non-polar compounds.
- [D] The carbonyl oxygen has non-bonding lone pairs that accept hydrogen bonds from water molecules, but long alkyl chains in hexan-2-one disrupt water's hydrogen bond network.

7 [EASY] Nucleophilic Attack Intermediate Structure — 9701/12/M/J/222/Q7**[1]**

In the reaction between propanone and HCN, what is the structure and geometry of the intermediate ion formed before protonation?

- [A] A planar carbocation intermediate.
- [B] A tetrahedral alkoxide intermediate, $(\text{CH}_3)_2\text{C}(\text{CN})\text{O}^-$, where carbon has transitioned to sp^3 hybridisation.
- [C] A linear free radical.
- [D] A five-coordinate transition state.

8 [HARD] Hydrolysis of 2-Hydroxynitriles — 9701/13/M/J/223/Q8**[1]**

What organic product is formed when the cyanohydrin 2-hydroxypropanenitrile is heated under reflux with dilute hydrochloric acid?

- [A] Propanoic acid only
- [B] Propan-1-amine
- [C] 2-hydroxypropanoic acid (lactic acid), $\text{CH}_3\text{-CH}(\text{OH})\text{-COOH}$, and ammonium chloride
- [D] Alanine

9 [EASY] Reduction of 2-Hydroxynitriles to Amino Alcohols — 9701/11/M/J/224/Q9**[1]**

What is produced when 2-hydroxypropanenitrile is reduced using LiAlH_4 in dry ether?

- [A] 2-aminopropanoic acid
- [B] 1-aminopropan-2-ol, $\text{CH}_3\text{-CH}(\text{OH})\text{-CH}_2\text{NH}_2$
- [C] Propane-1,2-diol
- [D] Propan-2-ol

10 [HARD] Reactivity: Aldehydes vs Ketones in Nucleophilic Addition — 9701/12/M/J/220/Q10**[1]**

Why are aliphatic aldehydes significantly more reactive towards nucleophilic addition than aliphatic ketones?

- [A] Aldehydes have only one electron-releasing alkyl group (+I effect) and one small hydrogen, making the carbonyl carbon more electron-deficient and sterically more accessible than in ketones.
- [B] Ketones have no dipole moment.
- [C] Aldehydes contain stronger covalent bonds.
- [D] Ketones are completely non-polar.

11 [EASY] Mechanism of Hydride Transfer from NaBH_4 — 9701/13/M/J/221/Q11**[1]**

In the reduction of a carbonyl compound by sodium borohydride, what is the acting nucleophile that attacks the carbonyl carbon?

- [A] A proton (H^+) from the solvent.
- [B] A borane radical ($\cdot\text{BH}_3$).
- [C] A sodium cation (Na^+).
- [D] A nucleophilic hydride ion (:H^-) transferred from the borohydride anion $[\text{BH}_4]^-$.

12 [HARD] Reduction using LiAlH_4 vs NaBH_4 — 9701/11/M/J/222/Q12**[1]**

Why can NaBH_4 be used safely in aqueous or alcoholic solvents, whereas LiAlH_4 must be used in strictly anhydrous ether solvents?

- [A] NaBH_4 is completely insoluble in water.
- [B] LiAlH_4 does not reduce carbonyl groups in ether.
- [C] LiAlH_4 reacts violently and explosively with water to evolve flammable hydrogen gas, whereas NaBH_4 is a milder reducing agent that hydrolyzes slowly in water.
- [D] Water decomposes NaBH_4 into chlorine gas.

13 [EASY] Addition of Sodium Hydrogensulfite (Bisulfite Addition) — 9701/12/M/J/223/Q13**[1]**

What crystalline product is formed when an aldehyde is shaken with a saturated aqueous solution of sodium hydrogensulfite (NaHSO_3)?

- [A] A white crystalline bisulfite addition adduct, $\text{R-CH(OH)-SO}_3^- \text{Na}^+$ (used to purify and isolate aldehydes).
- [B] A yellow precipitate of sulfur.
- [C] A solution of sodium sulfate.
- [D] An ester.

14 [HARD] Equilibrium in Carbonyl Hydration — 9701/13/M/J/224/Q14**[1]**

When methanal (formaldehyde) is dissolved in water, it exists almost 100% as the gem-diol methanediol, $\text{CH}_2(\text{OH})_2$. Why is methanal hydrated so completely compared to propanone?

- [A] Methanal reacts irreversibly with water to form methane.
- [B] Propanone is completely insoluble in water.
- [C] Methanal has no electron-donating alkyl groups and minimal steric hindrance, making its carbonyl carbon intensely δ^+ and favoring gem-diol equilibrium.
- [D] Methanediol contains ionic bonds.

15 [EASY] Addition of Grignard Reagents to Methanal — 9701/11/M/J/220/Q15**[1]**

What class of alcohol is synthesized when a Grignard reagent (R-Mg-X) reacts with methanal (HCHO) followed by acid hydrolysis?

- [A] Secondary alcohol
- [B] Primary alcohol, $\text{R-CH}_2\text{OH}$
- [C] Tertiary alcohol
- [D] Carboxylic acid

16 [HARD] Addition of Grignard Reagents to Other Aldehydes — 9701/12/M/J/221/Q16**[1]**

What class of alcohol is obtained when a Grignard reagent (R'-Mg-X) reacts with an aldehyde other than methanal (R-CHO)?

- [A] Primary alcohol
- [B] Tertiary alcohol
- [C] Ketone
- [D] Secondary alcohol, R-CH(OH)-R'

17 [EASY] Addition of Grignard Reagents to Ketones — 9701/13/M/J/222/Q17**[1]**

What class of alcohol is produced when a Grignard reagent reacts with a ketone (R-CO-R')?

- [A] Secondary alcohol
- [B] Primary alcohol
- [C] Tertiary alcohol, R-C(OH)(R')(R'')
- [D] Aldehyde

18 [HARD] Planar Geometry and Bond Angle of Carbonyls — 9701/11/M/J/223/Q18**[1]**

What is the C-C=O bond angle in propanone, and what is the hybridisation of the carbonyl oxygen atom?

- [A] Bond angle is approximately 120°; the oxygen atom is sp² hybridised with two non-bonding lone pairs in sp² orbitals.
- [B] Bond angle is 109.5°; oxygen is sp³ hybridised.
- [C] Bond angle is 180°; oxygen is sp hybridised.
- [D] Bond angle is 90°; oxygen is unhybridised.

19 [EASY] Polarity Comparison: C=O vs C=C — 9701/12/M/J/224/Q19**[1]**

Why does the C=O double bond undergo nucleophilic addition while the C=C double bond undergoes electrophilic addition?

- [A] The C=O bond has fewer electrons than C=C.
- [B] The C=C bond contains dative bonds.
- [C] Oxygen repels nucleophiles.
- [D] The C=O bond is strongly polarized with an electron-deficient carbon (C δ^+) that attracts nucleophiles; the C=C bond is non-polar with an exposed electron-rich pi cloud that attracts electrophiles.

20 [HARD] Reaction Coordinate of Cyanohydrin Formation — 9701/13/M/J/220/Q20**[1]**

In the reaction profile for the addition of HCN to an aldehyde, what is the rate-determining step?

- [A] The protonation of the alkoxide ion by HCN (Step 2).
- [B] The nucleophilic attack of the cyanide ion (:CN⁻) on the carbonyl carbon (Step 1).
- [C] The mixing of reagents.
- [D] The evaporation of HCN.

21 [EASY] Addition of Alcohols to Aldehydes (Hemiacetal Formation) — 9701/11/M/J/221/Q21**[1]**

What functional groups are present in a hemiacetal, formed by the addition of one mole of alcohol to an aldehyde?

- [A] Two ester groups.
- [B] An ether (-OR) and an alcohol (-OH) attached to the same carbon atom: R-CH(OH)(OR').
- [C] A ketone and an ether.
- [D] An alkene and an alcohol.

22 [HARD] Acetal Formation and Protection — 9701/12/M/J/222/Q22**[1]**

What is an acetal, and why are acetals used in organic synthesis?

- [A] An ester polymer used in nylon.
- [B] A salt of acetic acid.
- [C] A poisonous gas.
- [D] A compound with two ether groups on the same carbon (R-CH(OR')₂); used as reversible protecting groups for carbonyls against bases and reducing agents.

23 [EASY] Cyanohydrin Synthesis Safety — 9701/13/M/J/223/Q23**[1]**

Why is the reaction between carbonyl compounds and HCN carried out using a mixture of NaCN and dilute H₂SO₄ rather than liquid HCN?

- [A] HCN is an extremely volatile, highly toxic gas (boiling point 26 °C); generating it in situ at pH 8-9 provides safety and maintains the necessary concentration of :CN⁻ nucleophile.
- [B] Liquid HCN is completely non-reactive.
- [C] NaCN turns into harmless nitrogen gas.
- [D] H₂SO₄ acts as a reducing agent.

24 [HARD] Reactivity of Aromatic vs Aliphatic Aldehydes — 9701/11/M/J/224/Q24**[1]**

Why is benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$) less reactive towards nucleophilic addition than ethanal?

- [A] Benzaldehyde is completely insoluble in all solvents.
- [B] The benzene ring repels all nucleophiles by magnetic fields.
- [C] The carbonyl carbon's partial positive charge is reduced by delocalisation of pi electrons from the aromatic benzene ring into the carbonyl group.
- [D] Benzaldehyde contains no carbonyl group.

25 [EASY] Steric Hindrance in Ketone Addition — 9701/12/M/J/220/Q25**[1]**

Why does di-tert-butyl ketone, $((\text{CH}_3)_3\text{C})_2\text{C}=\text{O}$, fail to react with HCN even after heating for several days?

- [A] Two massive, bulky tertiary-butyl groups completely surround and shield the carbonyl carbon, sterically preventing the approach of cyanide ions.
- [B] It is a cyclic alkane.
- [C] The $\text{C}=\text{O}$ bond is too strong to break.
- [D] Cyanide ions decompose in the presence of tert-butyl groups.

26 [HARD] Preparation of Aldehydes from Primary Alcohols — 9701/13/M/J/221/Q26**[1]**

What apparatus is essential to obtain ethanal in good yield from the oxidation of ethanol by acidified potassium dichromate(VI)?

- [A] Reflux apparatus with continuous boiling.
- [B] Buchner filtration funnel.
- [C] Separating funnel with drying agent.
- [D] Distillation apparatus with immediate collection of distillate (to separate volatile ethanal before it oxidizes to ethanoic acid).

27 [EASY] Preparation of Ketones from Secondary Alcohols — 9701/11/M/J/222/Q27**[1]**

Why does the preparation of propanone from propan-2-ol by acidified $\text{K}_2\text{Cr}_2\text{O}_7$ use heating under reflux rather than distillation?

- [A] Propanone boils at a higher temperature than propan-2-ol.
- [B] Ketones cannot be oxidized further under these conditions, so prolonged heating under reflux ensures complete conversion of the alcohol.
- [C] Refluxing removes unreacted acid.
- [D] Propanone explodes if distilled.

28 [HARD] Detection of Carbonyls in Atmospheric Smog — 9701/12/M/J/223/Q28**[1]**

Why are low-molecular-weight aldehydes (such as methanal and ethanal) hazardous pollutants in urban photochemical smog?

- [A] They destroy asphalt roads.
- [B] They cause immediate freezing of water vapor.
- [C] They are powerful eye and respiratory tract irritants, toxic to lung tissue, and precursors to peroxyacetyl nitrate (PAN).
- [D] They deplete nitrogen from the air.

29 [EASY] Equilibrium Constant for HCN Addition — 9701/13/M/J/224/Q29**[1]**

How does the equilibrium constant (K) for the addition of HCN vary across the carbonyl series: methanal, ethanal, propanone?

- [A] Propanone > Ethanal > Methanal
- [B] Methanal > Ethanal > Propanone
- [C] All three have identical equilibrium constants.
- [D] Propanone does not reach equilibrium.

30 [HARD] Addition of Water (Hydration) of Hexafluoroacetone — 9701/11/M/J/220/Q30**[1]**

Hexafluoroacetone, $(\text{CF}_3)_2\text{C}=\text{O}$, reacts exothermically with water to form a stable crystalline gem-diol, $(\text{CF}_3)_2\text{C}(\text{OH})_2$. What explains this unusual stability?

- [A] The six intensely electronegative fluorine atoms exert an extreme electron-withdrawing ($-I$) inductive effect, making the carbonyl carbon massively electron-deficient.
- [B] Fluorine atoms form covalent bonds with water.
- [C] Hexafluoroacetone is an ionic compound.
- [D] The gem-diol is aromatic.

31 [EASY] Reduction of Unsaturated Carbonyls with NaBH_4 — 9701/12/M/J/221/Q31**[1]**

When but-2-enal (crotonaldehyde, $\text{CH}_3\text{-CH=CH-CHO}$) is treated with NaBH_4 in ethanol, what is the organic product?

- [A] Butan-1-ol (both bonds reduced).
- [B] But-2-enoic acid.
- [C] Butane.
- [D] But-2-en-1-ol, $\text{CH}_3\text{-CH=CH-CH}_2\text{OH}$ (the $\text{C}=\text{O}$ is reduced while the $\text{C}=\text{C}$ double bond remains untouched).

32 [HARD] Reduction of Unsaturated Carbonyls with H_2/Ni — 9701/13/M/J/222/Q32**[1]**

When but-2-enal ($\text{CH}_3\text{-CH=CH-CHO}$) is reacted with excess hydrogen gas over a nickel catalyst at 150°C , what product is formed?

- [A] But-2-en-1-ol.
- [B] Butanal.
- [C] Butan-1-ol, $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{OH}$ (both the $\text{C}=\text{C}$ and $\text{C}=\text{O}$ bonds are completely hydrogenated).
- [D] Butane.

33 [EASY] Odor of Carbonyl Compounds — 9701/11/M/J/223/Q33**[1]**

How does the odor of aldehydes change as the carbon chain length increases from C_1 to C_{10} ?

- [A] Small aldehydes ($\text{C}_1\text{-C}_3$) have sharp, pungent, unpleasant choking smells; medium-chain aldehydes ($\text{C}_8\text{-C}_{10}$) have pleasant floral, citrus, or fruity fragrances used in perfumes.
- [B] All aldehydes smell of rotten eggs.
- [C] Aldehydes have no odor.
- [D] Higher aldehydes smell of ammonia.

34 [HARD] Reaction of Carbonyls with Hydroxylamine — 9701/12/M/J/224/Q34**[1]**

What type of compound is formed when an aldehyde reacts with hydroxylamine (NH_2OH) with elimination of water?

- [A] A nitrile
- [B] An amide
- [C] An oxime, R-CH=N-OH (condensation reaction)
- [D] An amine

35 [EASY] Reaction of Carbonyls with Hydrazine — 9701/13/M/J/220/Q35**[1]**

What product is formed when propanone reacts with hydrazine (NH_2NH_2)?

- [A] An amino acid
- [B] A hydrazone, $(\text{CH}_3)_2\text{C=N-NH}_2$, and water
- [C] Ethanenitrile
- [D] Propan-2-amine

36 [HARD] Wolff-Kishner Reduction Overview — 9701/11/M/J/221/Q36**[1]**

What transformation occurs when a hydrazone of a ketone is heated with strong base (KOH) at 200 °C?

- [A] The ketone is oxidized to an ester.
- [B] The molecule undergoes cyclization.
- [C] The ketone is chlorinated.
- [D] The carbonyl group ($>\text{C}=\text{O}$) is completely reduced to a methylene group ($-\text{CH}_2-$), releasing nitrogen gas (N_2).

37 [EASY] Clemmensen Reduction Comparison — 9701/12/M/J/222/Q37**[1]**

Which reagents achieve the reduction of a ketone directly to an alkane under strongly acidic conditions?

- [A] Sodium borohydride in ethanol
- [B] Lithium aluminium hydride in ether
- [C] Zinc amalgam (Zn/Hg) and concentrated hydrochloric acid (Clemmensen reduction)
- [D] Acidified potassium dichromate

38 [HARD] Keto-Enol Tautomerism — 9701/13/M/J/223/Q38**[1]**

What is keto-enol tautomerism in carbonyl compounds such as propanone?

- [A] A dynamic equilibrium between a keto isomer ($\text{CH}_3\text{-CO-CH}_3$) and an enol isomer ($\text{CH}_2=\text{C}(\text{OH})\text{-CH}_3$) involving the migration of a proton and double bond.
- [B] A pair of optical enantiomers.
- [C] A mixture of cis and trans isomers.
- [D] A transition state during reduction.

39 [EASY] Alpha-Hydrogen Acidity in Carbonyls — 9701/11/M/J/224/Q39**[1]**

Why are hydrogen atoms on the carbon adjacent to a carbonyl group (alpha-hydrogens) significantly more acidic ($\text{pK}_a \sim 20$) than alkane hydrogens ($\text{pK}_a \sim 50$)?

- [A] Alpha-hydrogens are bonded to oxygen.
- [B] The carbonyl group repels electrons into space.
- [C] Alpha-carbons carry a full negative charge.
- [D] Deprotonation generates an enolate carbanion that is stabilized by resonance delocalisation of the negative charge onto the electronegative oxygen atom.

40 [HARD] Aldol Condensation Overview — 9701/12/M/J/220/Q40**[1]**

What happens when ethanal is treated with dilute aqueous sodium hydroxide at room temperature?

- [A] Ethanal is oxidized to ethanoic acid.
- [B] Two molecules of ethanal combine via an aldol addition to form 3-hydroxybutanal (an aldol).
- [C] Ethanal is cleaved into methane and CO_2 .
- [D] Ethanal polymerises into polystyrene.

41 [EASY] Addition of Sodium Cyanide to Symmetrical Ketones — 9701/13/M/J/221/Q41**[1]**

When propanone (CH_3COCH_3) reacts with HCN/NaCN , why is the resulting product 2-hydroxy-2-methylpropanenitrile OPTICALLY INACTIVE?

- [A] The product is a racemic mixture.
- [B] The product molecule contains two identical methyl groups attached to C_2 , so it has no chiral center.
- [C] The reaction proceeds with complete retention.
- [D] Propanone is a solid.

42 [HARD] Deducing Carbonyl Structure from Reduction Product — 9701/11/M/J/222/Q42**[1]**

Reduction of an unknown carbonyl compound Z with NaBH_4 gives 3-methylbutan-2-ol. What is the identity of compound Z?

- [A] 3-methylbutanal
- [B] Pentan-2-one
- [C] 2-methylbutanal
- [D] 3-methylbutan-2-one, $(\text{CH}_3)_2\text{CH}-\text{CO}-\text{CH}_3$

43 [EASY] Deducing Carbonyl Structure from Primary Alcohol Product — 9701/12/M/J/223/Q43**[1]**

Reduction of carbonyl compound W with NaBH_4 produces 2,2-dimethylpropan-1-ol. What is compound W?

- [A] 2,2-dimethylpropanal, $(\text{CH}_3)_3\text{C}-\text{CHO}$
- [B] 3-methylbutanal
- [C] 2,2-dimethylpropan-2-ol
- [D] Pentan-3-one

44 [HARD] Carboxylic Acid Derivatives: Acyl Chlorides vs Carbonyls — 9701/13/M/J/224/Q44**[1]**

Why do acyl chlorides (RCOCl) undergo nucleophilic substitution while aldehydes (RCHO) undergo nucleophilic addition?

- [A] Aldehydes contain no polar bonds.
- [B] Acyl chlorides have no carbonyl groups.
- [C] Chloride ($:\text{Cl}^-$) is an excellent leaving group that can be eliminated after initial addition, whereas hydride ($:\text{H}^-$) is an extremely poor leaving group that cannot be expelled.
- [D] Chlorine atoms attract protons.

45 [EASY] Hydrolysis of Cyanohydrin Reaction Equation — 9701/11/M/J/220/Q45**[1]**

What is the balanced equation for the acid hydrolysis of 2-hydroxypropanenitrile?

- [A] $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$
- [B] $\text{CH}_3\text{CH}(\text{OH})\text{CN} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CONH}_2$
- [C] $\text{CH}_3\text{CH}(\text{OH})\text{CN} + \text{HCl} \rightarrow \text{CH}_3\text{CH}(\text{Cl})\text{CN} + \text{H}_2\text{O}$
- [D] $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 2\text{H}_2 \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{NH}_2$

46 [HARD] Cyanohydrin Formation with Benzaldehyde — 9701/12/M/J/221/Q46**[1]**

What is the product when benzaldehyde reacts with HCN in the presence of NaCN ?

- [A] Benzoic acid
- [B] Phenylmethanol
- [C] Benzylamine
- [D] 2-hydroxy-2-phenylethanenitrile (mandelonitrile), $\text{C}_6\text{H}_5-\text{CH}(\text{OH})-\text{CN}$

47 [EASY] Effect of pH on HCN Addition Rate — 9701/13/M/J/222/Q47**[1]**

Why is the rate of addition of HCN to propanone maximum at pH 8–9, decreasing at both lower and higher pH?

- [A] At pH 8 propanone boils.
- [B] At pH < 7, there are too few $:\text{CN}^-$ nucleophiles to initiate attack; at pH > 10, there is insufficient undissociated HCN to protonate the intermediate alkoxide.
- [C] At low pH the $\text{C}=\text{O}$ bond breaks.
- [D] At high pH sodium ions destroy the carbonyl.

48 [HARD] Acid Catalysis in Carbonyl Addition — 9701/11/M/J/223/Q48

[1]

Why does trace acid catalyze the addition of weak nucleophiles to carbonyl compounds?

- [A] Acid increases the concentration of hydroxide.
- [B] Acid removes the pi bond completely.
- [C] Protonation of the carbonyl oxygen atom ($>\text{C}=\text{O}+\text{H}$) greatly increases the partial positive charge on carbon, making it much more attractive to weak nucleophiles.
- [D] Acid acts as an oxidizing agent.

49 [EASY] Chloral Hydrate Stability — 9701/12/M/J/224/Q49

[1]

Why does trichloroethanal (chloral, CCl_3CHO) form a remarkably stable crystalline gem-diol, chloral hydrate ($\text{CCl}_3\text{CH}(\text{OH})_2$)?

- [A] Chloral hydrate is an ionic crystal of sodium.
- [B] Three strongly electron-withdrawing chlorine atoms withdraw electron density from the carbonyl carbon, combined with intramolecular hydrogen bonding in the gem-diol.
- [C] The C-Cl bonds are destroyed by water.
- [D] Chloral is an alkene.

50 [HARD] Preparation of Aldehydes via Dehydrogenation — 9701/13/M/J/220/Q50

[1]

What reaction occurs when ethanol vapor is passed over heated copper metal at 300°C in the absence of air?

- [A] Dehydrogenation to produce ethanal (CH_3CHO) and hydrogen gas (H_2).
- [B] Dehydration to ethene and steam.
- [C] Combustion to carbon dioxide.
- [D] Reduction to ethane.

51 [EASY] Formaldehyde Polymerisation in Aqueous Solution — 9701/11/M/J/221/Q51

[1]

When aqueous methanal (formalin) is concentrated, what solid white polymer is formed?

- [A] Polyethene
- [B] Polyvinyl chloride
- [C] Nylon-6,6
- [D] Paraformaldehyde, $\text{HO}-(\text{CH}_2\text{O})_n-\text{H}$

52 [HARD] Addition of HCN to 2-chloropropanal — 9701/12/M/J/222/Q52

[1]

When 2-chloropropanal ($\text{CH}_3-\text{CH}(\text{Cl})-\text{CHO}$) reacts with HCN/NaCN , how many chiral centers are present in the product?

- [A] 1 chiral center
- [B] 0 chiral centers
- [C] 2 chiral centers
- [D] 3 chiral centers

53 [EASY] Reduction Stoichiometry with LiAlH_4 — 9701/13/M/J/223/Q53

[1]

How many moles of ethanol are produced when one mole of ethyl ethanoate is completely reduced by LiAlH_4 ?

- [A] 2 moles of ethanol (one from the acyl portion and one from the alkoxy portion)
- [B] 1 mole of ethanol
- [C] 0.5 mole of ethanol
- [D] 4 moles of ethanol

54 [HARD] Carbonyl Dipole Moments Comparison — 9701/11/M/J/224/Q54

[1]

Why is the dipole moment of propanone (2.88 D) significantly higher than that of 1-chloropropane (2.04 D) despite chlorine being more electronegative than oxygen?

- [A] Carbonyls have ionic bonds.
- [B] 1-chloropropane is planar.
- [C] The C=O group contains an easily polarizable pi bond in addition to a sigma bond, allowing greater electronic displacement toward oxygen.
- [D] Propanone contains hydrogen bonds.

55 [EASY] Hydride Attack Trajectory (Bürgi-Dunitz Angle) — 9701/12/M/J/220/Q55

[1]

At what approximate angle does a nucleophile attack the planar carbonyl carbon atom during nucleophilic addition?

- [A] Directly along the bond axis at 180°
- [B] Approximately 107° to the C=O bond axis (the Bürgi-Dunitz angle)
- [C] Perpendicular to the plane at 90°
- [D] Along the C-C bond at 120°

SUBTOPIC 17.2 — DIAGNOSTIC TESTING & THE IODOFORM REACTION (Q56 – Q100)**56** [EASY] Diagnostic Test for Carbonyl Group with 2,4-DNPH — 9701/11/M/J/23/Q27

[1]

What observation confirms the presence of an aldehyde or ketone when Brady's reagent (2,4-dinitrophenylhydrazine) is added to an unknown liquid?

- [A] Evolution of bubbles of carbon dioxide gas.
- [B] Formation of a silver mirror on the glass walls.
- [C] A color change from orange to bright green.
- [D] Immediate formation of a vivid yellow, orange, or red crystalline precipitate.

57 [HARD] Identification of Specific Carbonyls via 2,4-DNPH Derivatives — 9701/12/M/J/23/Q27

[1]

How can the 2,4-dinitrophenylhydrazone precipitate be used to definitively identify which specific aldehyde or ketone was present in an unknown sample?

- [A] Dissolve the precipitate in acid and titrate with sodium hydroxide.
- [B] Measure the boiling point of the wet precipitate.
- [C] Filter and purify the precipitate by recrystallisation, dry it, measure its sharp melting point, and compare it with tables of known derivative melting points.
- [D] Count the number of crystals under an optical microscope.

58 [EASY] Tollens' Reagent Test for Aldehydes — 9701/13/M/J/23/Q27

[1]

What is Tollens' reagent, and what is observed when it is warmed with an aldehyde versus a ketone?

- [A] Ammoniacal silver nitrate, containing $[\text{Ag}(\text{NH}_3)_2]^+$; aldehydes produce a reflective silver mirror (or grey precipitate), while ketones show no reaction.
- [B] Acidified potassium dichromate; aldehydes turn green, ketones turn purple.
- [C] Copper(II) sulfate in alkali; aldehydes form a blue precipitate.
- [D] Iodine in sodium hydroxide; both form a yellow precipitate.

59 [EASY] Fehling's Solution Test for Aldehydes — 9701/11/O/N/23/Q27

[1]

What observation occurs when ethanal is heated with Fehling's solution (an alkaline solution of copper(II) tartrate)?

- [A] The solution turns colorless with evolution of hydrogen gas.
- [B] A shiny mirror of copper metal coats the flask.
- [C] A pale yellow crystalline precipitate is formed.
- [D] The deep blue solution forms a dense brick-red precipitate of copper(I) oxide, Cu_2O .

60 [HARD] The Tri-iodomethane (Iodoform) Reaction for Carbonyls — 9701/11/M/J/220/Q60

[1]

Which structural grouping is required for an aldehyde or ketone to give a positive tri-iodomethane (iodoform) test?

- [A] An aldehyde group, -CHO
- [B] A methyl carbonyl group, $\text{CH}_3\text{-C(=O)-}$
- [C] A secondary alcohol group, -CH(OH)-
- [D] An ester group, -COOCH_3

61 [EASY] Only Aldehyde to Give Positive Iodoform Test — 9701/12/M/J/221/Q61

[1]

Why is ethanal (CH_3CHO) the ONLY aldehyde that gives a yellow precipitate with alkaline aqueous iodine?

- [A] It has the lowest boiling point of all aldehydes.
- [B] It is the only aldehyde with a methyl group bonded directly to the carbonyl carbon ($\text{CH}_3\text{-C(=O)-H}$).
- [C] It is the only aldehyde that is soluble in water.
- [D] It is the only aldehyde that can be reduced to an alcohol.

62 [HARD] Identification of Methyl Ketones via Iodoform — 9701/13/M/J/222/Q62

[1]

Which of the following ketones produces a yellow precipitate of tri-iodomethane when warmed with I_2 and aqueous NaOH ?

- [A] Pentan-3-one, $\text{CH}_3\text{CH}_2\text{-CO-CH}_2\text{CH}_3$
- [B] Cyclohexanone
- [C] Hexan-3-one, $\text{CH}_3\text{CH}_2\text{-CO-CH}_2\text{CH}_2\text{CH}_3$
- [D] Pentan-2-one, $\text{CH}_3\text{-CO-CH}_2\text{CH}_2\text{CH}_3$

63 [EASY] Iodoform Cleavage Byproduct — 9701/11/M/J/223/Q63

[1]

When propanone reacts with alkaline iodine to form tri-iodomethane (CHI_3), what is the other organic product formed in solution?

- [A] Sodium ethanoate, CH_3COONa
- [B] Sodium methanoate, HCOONa
- [C] Sodium propanoate, $\text{CH}_3\text{CH}_2\text{COONa}$
- [D] Methanol, CH_3OH

64 [HARD] Tollens' Reagent Redox Half-Equations — 9701/12/M/J/224/Q64

[1]

In the reaction between an aldehyde and Tollens' reagent, what are the oxidation and reduction half-equations?

- [A] Oxidation: $\text{Ag}^+ \rightarrow \text{Ag}^{2+} + \text{e}^-$; Reduction: $\text{RCHO} + 2\text{e}^- \rightarrow \text{RCH}_2\text{OH}$
- [B] Oxidation: $\text{RCHO} \rightarrow \text{RH} + \text{CO} + \text{e}^-$; Reduction: $\text{NH}_3 + \text{e}^- \rightarrow \text{NH}_2^-$
- [C] Oxidation: $\text{RCHO} + 3\text{OH}^- \rightarrow \text{RCOO}^- + 2\text{H}_2\text{O} + 2\text{e}^-$; Reduction: $[\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^- \rightarrow \text{Ag(s)} + 2\text{NH}_3$
- [D] Oxidation: $\text{OH}^- \rightarrow \text{O}_2 + \text{H}^+ + \text{e}^-$; Reduction: $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$

65 [EASY] Fehling's Test Limitation with Aromatic Aldehydes — 9701/13/M/J/220/Q65

[1]

Why does benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$) give a silver mirror with Tollens' reagent but fail to reduce Fehling's solution?

- [A] Fehling's solution is a weaker oxidizing agent than Tollens' reagent and cannot overcome the resonance stabilization of aromatic aldehydes.
- [B] Benzaldehyde is completely insoluble in water and decomposes Fehling's.
- [C] Benzaldehyde is a ketone.
- [D] Copper ions precipitate benzoic acid.

66 [HARD] Acidified Potassium Dichromate Oxidation of Aldehydes — 9701/11/M/J/221/Q66**[1]**

What is observed when ethanal is warmed with acidified potassium dichromate(VI)?

- [A] The orange solution turns purple.
- [B] A silver mirror forms on the glass.
- [C] Vigorous effervescence of hydrogen gas.
- [D] The orange solution turns green ($\text{Cr}_2\text{O}_7^{2-}$ reduced to Cr^{3+}) as ethanal is oxidized to ethanoic acid.

67 [EASY] Resistance of Ketones to Acidified Dichromate — 9701/12/M/J/222/Q67**[1]**

What happens when propanone is heated with acidified potassium dichromate(VI)?

- [A] The solution turns green.
- [B] No reaction occurs; the solution remains orange.
- [C] A brick-red precipitate forms.
- [D] A silver mirror forms.

68 [HARD] Differentiating Propanal and Propanone — 9701/13/M/J/223/Q68**[1]**

Which test CANNOT be used to distinguish propanal from propanone?

- [A] Tollens' reagent (propanal forms a silver mirror; propanone does not)
- [B] Fehling's solution (propanal forms a brick-red precipitate; propanone does not)
- [C] 2,4-dinitrophenylhydrazine (both give an orange precipitate)
- [D] Acidified potassium dichromate (propanal turns it green; propanone does not)

69 [EASY] Distinguishing Ethanal and Propanal — 9701/11/M/J/224/Q69**[1]**

Which chemical test cleanly differentiates ethanal from propanal?

- [A] Tollens' reagent (both form silver mirrors).
- [B] The tri-iodomethane (iodoform) test: ethanal forms a yellow precipitate of CHI_3 ; propanal gives no precipitate.
- [C] Fehling's solution (both form red precipitates).
- [D] 2,4-DNPH (both form orange precipitates).

70 [HARD] Distinguishing Propanone and Pentan-3-one — 9701/12/M/J/220/Q70**[1]**

Which single test differentiates propanone from pentan-3-one?

- [A] Alkaline aqueous iodine (iodoform test): propanone forms a yellow precipitate of CHI_3 ; pentan-3-one gives no precipitate.
- [B] 2,4-DNPH (both form orange precipitates).
- [C] Tollens' reagent (neither reacts).
- [D] Fehling's solution (neither reacts).

71 [EASY] Mechanism of 2,4-DNPH Condensation — 9701/13/M/J/221/Q71**[1]**

What type of overall reaction occurs between a carbonyl compound and 2,4-dinitrophenylhydrazine?

- [A] Electrophilic substitution.
- [B] Free radical addition.
- [C] Nucleophilic substitution of oxygen.
- [D] Addition-elimination (condensation), with the loss of a water molecule.

72 [HARD] Structure of 2,4-DNPH — 9701/11/M/J/222/Q72**[1]**

What is the structural formula of 2,4-dinitrophenylhydrazine?

- [A] A phenyl ring with two amino groups and four nitro groups.
- [B] A cyclohexyl ring with an amine and carboxylic acid.
- [C] A benzene ring substituted with a hydrazine group ($-\text{NHNH}_2$) at C1, and nitro groups ($-\text{NO}_2$) at C2 and C4.
- [D] An aliphatic diamine.

73 [EASY] Why 2,4-DNPH is Used Instead of Hydrazine — 9701/12/M/J/223/Q73**[1]**

Why is 2,4-dinitrophenylhydrazine preferred over simple hydrazine (NH_2NH_2) for identifying carbonyl compounds?

- [A] The 2,4-dinitrophenylhydrazones have higher molecular masses and aromatic pi systems, forming highly crystalline, easily purified solids with sharp melting points.
- [B] Hydrazine does not react with aldehydes.
- [C] Hydrazine is a colored dye that stains glassware.
- [D] Hydrazine only reacts at 500 °C.

74 [HARD] Recrystallisation Procedure for 2,4-DNPH Derivatives — 9701/13/M/J/224/Q74**[1]**

What is the proper laboratory procedure for recrystallising a crude 2,4-dinitrophenylhydrazone precipitate?

- [A] Boil until all solvent evaporates completely.
- [B] Dissolve in cold water and add concentrated sulfuric acid.
- [C] Dissolve in the minimum volume of hot solvent, filter hot to remove insoluble impurities, cool slowly to form pure crystals, filter under suction, and wash with cold solvent.
- [D] Heat to its melting point and let it solidify.

75 [EASY] Melting Point Determination Technique — 9701/11/M/J/220/Q75**[1]**

How is the melting point of a dried 2,4-dinitrophenylhydrazone measured accurately in a capillary tube?

- [A] Heat rapidly with a Bunsen burner flame directly on the crystals.
- [B] Pack a small amount in a thin glass capillary, heat slowly in an oil bath or melting point apparatus (1–2 °C per minute near the melting point), and record the temperature range over which it liquefies.
- [C] Dissolve in ethanol and use a thermometer.
- [D] Measure the temperature at which it catches fire.

76 [HARD] Preparation of Tollens' Reagent in the Laboratory — 9701/12/M/J/221/Q76**[1]**

How is Tollens' reagent freshly prepared immediately before testing in the laboratory?

- [A] Mix silver nitrate with concentrated hydrochloric acid.
- [B] Dissolve silver metal in boiling nitric acid.
- [C] Add potassium iodide to silver nitrate.
- [D] Add dilute NaOH to aqueous AgNO_3 to precipitate brown Ag_2O , then add dilute aqueous NH_3 dropwise until the precipitate just dissolves.

77 [EASY] Explosion Hazard of Tollens' Reagent — 9701/13/M/J/222/Q77**[1]**

Why must any leftover Tollens' reagent be rinsed down the sink with copious water and acidified immediately after use?

- [A] It releases toxic fluorine gas upon standing.
- [B] It corrodes glass into liquid silica.
- [C] Upon standing, it can decompose into highly sensitive, contact-explosive silver nitride (Ag_3N , fulminating silver).
- [D] It absorbs atmospheric radon.

78 [HARD] Testing Cleanness of Glassware for Silver Mirror — 9701/11/M/J/223/Q78**[1]**

Why must the test tube be thoroughly degreased and cleaned with hot aqueous NaOH before performing the Tollens' test?

- [A] Any grease or dirt prevents the silver atoms from depositing as a smooth, uniform reflective mirror on the glass, resulting instead in an unsightly dull grey/black precipitate.
- [B] Grease oxidizes the aldehyde prematurely.
- [C] Grease dissolves silver metal.
- [D] Dirty glass neutralizes the ammonia complex.

79 [EASY] Structure of Copper(I) Oxide Precipitate — 9701/12/M/J/224/Q79**[1]**

What is the chemical formula, oxidation state of copper, and visual color of the precipitate formed in a positive Fehling's test?

- [A] CuO ; copper is in the +2 oxidation state; black precipitate.
- [B] Cu ; copper is in the 0 oxidation state; bronze solid.
- [C] $\text{Cu}(\text{OH})_2$; copper is in the +2 oxidation state; pale blue gelatinous solid.
- [D] Cu_2O ; copper is in the +1 oxidation state; brick-red precipitate.

80 [HARD] Benedict's Solution vs Fehling's Solution — 9701/13/M/J/220/Q80**[1]**

How does Benedict's solution differ from Fehling's solution, and what is its practical advantage?

- [A] Benedict's solution contains gold ions instead of copper.
- [B] Benedict's solution uses sodium citrate and sodium carbonate instead of tartrate and NaOH , making it a single stable solution with a longer shelf life.
- [C] Benedict's solution tests exclusively for ketones.
- [D] Benedict's solution is acidic.

81 [EASY] Oxidation of Methanal with Fehling's — 9701/11/M/J/221/Q81**[1]**

How many moles of Cu_2O precipitate are formed when one mole of methanal (HCHO) is completely oxidized by excess Fehling's solution?

- [A] 1 mole of Cu_2O
- [B] 2 moles of Cu_2O (because methanal can be oxidized in two successive stages to CO_2)
- [C] 0.5 mole of Cu_2O
- [D] 4 moles of Cu_2O

82 [HARD] Distinguishing Secondary Alcohol from Methyl Ketone — 9701/12/M/J/222/Q82**[1]**

Both butan-2-ol and butanone give a yellow precipitate with alkaline iodine. Which reagent cleanly distinguishes between them?

- [A] 2,4-DNPH (both give yellow precipitates).
- [B] Alkaline iodine.
- [C] Sodium hydroxide solution.
- [D] Acidified potassium dichromate(VI): butan-2-ol turns the orange solution green; butanone causes no color change.

83 [EASY] Combustion of Tri-iodomethane — 9701/13/M/J/223/Q83**[1]**

What is the molecular formula and approximate relative molecular mass of tri-iodomethane (iodoform)?

- [A] CHI_3 , $M_r = 393.7$
- [B] CH_3I , $M_r = 141.9$
- [C] CI_4 , $M_r = 519.6$
- [D] CH_2I_2 , $M_r = 267.8$

84 [HARD] Identification of Carbonyl with $\text{C}_5\text{H}_{10}\text{O}$ — 9701/11/M/J/224/Q84**[1]**

A carbonyl compound of formula $\text{C}_5\text{H}_{10}\text{O}$ forms an orange precipitate with 2,4-DNPH, does NOT react with Tollens' reagent, and gives a yellow precipitate with alkaline iodine. What is the compound?

- [A] Pentan-3-one
- [B] Pentanal
- [C] Pentan-2-one or 3-methylbutan-2-one
- [D] 2-methylbutanal

85 [EASY] Reaction of 2,4-DNPH with Alcohols and Carboxylic Acids — 9701/12/M/J/220/Q85**[1]**

Why do alcohols (ROH), carboxylic acids (RCOOH), and esters (RCOOR') NOT form a precipitate with 2,4-DNPH?

- [A] Their hydroxyl/alkoxy oxygen atoms donate electron density to the carbonyl carbon via resonance (+M), drastically reducing electrophilicity and preventing condensation.
- [B] They do not dissolve in water.
- [C] They decompose 2,4-DNPH into nitrogen gas.
- [D] They are solid polymers.

86 [HARD] Color Variations in 2,4-DNPH Derivatives — 9701/13/M/J/221/Q86**[1]**

Why do conjugated or aromatic carbonyls (such as benzaldehyde or cinnamaldehyde) yield deep red 2,4-DNPH precipitates, while saturated aliphatic carbonyls (such as propanal) yield yellow precipitates?

- [A] Aromatic compounds are hotter.
- [B] Red precipitates contain iron impurities.
- [C] Yellow precipitates contain unreacted sulfur.
- [D] Extended conjugation between the aromatic ring, C=N double bond, and dinitrophenyl group narrows the HOMO-LUMO energy gap, shifting absorption to longer wavelengths (red).

87 [EASY] Tri-iodomethane Test Reaction Conditions — 9701/11/M/J/222/Q87**[1]**

Why is the tri-iodomethane test conducted at gentle warmth (50–60 °C) rather than boiling?

- [A] Boiling explodes the iodine crystals.
- [B] Boiling would vaporise volatile carbonyl reactants and decompose the hypoiodite (IO⁻) reagent into iodate (IO₃⁻).
- [C] Tri-iodomethane is only solid above 100 °C.
- [D] Sodium hydroxide decomposes at 80 °C.

88 [HARD] Oxidation of Acetophenone — 9701/12/M/J/223/Q88**[1]**

What is the product when phenylethanone (acetophenone, C₆H₅-CO-CH₃) is warmed with alkaline aqueous iodine?

- [A] Benzoic acid and iodomethane.
- [B] Benzophenone.
- [C] A yellow precipitate of tri-iodomethane (CHI₃) and a solution of sodium benzoate (C₆H₅COONa).
- [D] No reaction occurs because it is aromatic.

89 [EASY] Silver Mirror Regeneration of Ag — 9701/13/M/J/224/Q89**[1]**

In chemical recycling of laboratory waste, how can silver metal deposited in Tollens' test tubes be reclaimed?

- [A] Scraping with a metal spatula.
- [B] Dissolving the silver mirror in concentrated nitric acid to regenerate aqueous silver nitrate: $3\text{Ag} + 4\text{HNO}_3 \rightarrow 3\text{AgNO}_3 + \text{NO} + 2\text{H}_2\text{O}$.
- [C] Heating in an oven at 200 °C.
- [D] Washing with ethanol.

90 [HARD] Qualitative Analysis Flowchart: C₃ Compounds — 9701/11/M/J/220/Q90**[1]**

Three unlabelled bottles contain propan-1-ol, propan-2-ol, and propanal.
Which test identifies propanal immediately from the other two?

- [A] Tollens' reagent (only propanal forms a silver mirror; neither alcohol reacts).
- [B] Acidified potassium dichromate (all three turn green).
- [C] Reaction with sodium metal (only propanal does not react).
- [D] The iodoform test.

91 [EASY] Qualitative Analysis Flowchart: Distinction of Alcohols — 9701/12/M/J/221/Q91**[1]**

After identifying propanal using Tollens' reagent, how can the remaining propan-1-ol and propan-2-ol be distinguished?

- [A] Reaction with PCl_5 .
- [B] Reaction with sodium metal.
- [C] Testing with litmus paper.
- [D] The tri-iodomethane test: propan-2-ol produces a yellow precipitate of CHI_3 , while propan-1-ol gives no precipitate.

92 [HARD] Deducing Carbonyl Structure from Quantitative Iodoform — 9701/13/M/J/222/Q92**[1]**

0.010 mol of a methyl ketone is reacted with alkaline iodine. What mass of tri-iodomethane (CHI_3 , $M_r = 393.7$) is precipitated?

- [A] 1.97 g of CHI_3
- [B] 7.88 g of CHI_3
- [C] 3.94 g of CHI_3
- [D] 0.39 g of CHI_3

93 [EASY] Role of Tartrate in Fehling's Solution — 9701/11/M/J/223/Q93**[1]**

What is the vital chemical function of sodium potassium tartrate in Fehling's B solution?

- [A] It acts as a bidentate chelating ligand that binds Cu^{2+} ions in alkaline solution, preventing them from precipitating as insoluble copper(II) hydroxide, $\text{Cu}(\text{OH})_2$.
- [B] It oxidizes the aldehyde directly.
- [C] It acts as an indicator.
- [D] It neutralizes acid.

94 [HARD] Role of Ammonia in Tollens' Reagent — 9701/12/M/J/224/Q94**[1]**

Why is aqueous ammonia added to silver nitrate in Tollens' reagent?

- [A] It acts as the oxidizing agent.
- [B] It neutralizes the aldehyde.
- [C] It complexes Ag^+ into the soluble diamminesilver(I) cation, $[\text{Ag}(\text{NH}_3)_2]^+$, preventing Ag^+ from precipitating as insoluble brown Ag_2O in alkaline medium.
- [D] It dissolves the silver mirror.

95 [EASY] Identification of Carbonyl with $M_r = 72$ — 9701/13/M/J/220/Q95**[1]**

A carbonyl compound has relative molecular mass $M_r = 72.0$.

- It gives an orange precipitate with 2,4-DNPH.
- It gives a brick-red precipitate with Fehling's solution.
- It does NOT give a yellow precipitate with alkaline iodine.

What is the compound?

- [A] Butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$
- [B] Butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- [C] 2-methylpropanal, $(\text{CH}_3)_2\text{CHCHO}$
- [D] Pentan-2-one

96 [HARD] Differentiating Butanal and 2-methylpropanal — 9701/11/M/J/221/Q96**[1]**

How can structural isomers butanal and 2-methylpropanal be definitively distinguished experimentally?

- [A] Using Fehling's solution.
- [B] Using Tollens' reagent.
- [C] Using acidified potassium dichromate.
- [D] By preparing their 2,4-dinitrophenylhydrazone derivatives, purifying by recrystallisation, and measuring their distinct melting points.

97 [EASY] Testing for Carbonyls with Schiff's Reagent — 9701/12/M/J/222/Q97**[1]**

What is Schiff's reagent, and what color change occurs when an aldehyde is added?

- [A] A solution of potassium permanganate that turns green.
- [B] A copper sulfate solution that turns yellow.
- [C] A solution of rosaniline hydrochloride decolorized by sulfur dioxide; aldehydes restore a vivid magenta/purple color without heating.
- [D] An iodine solution that turns blue.

98 [HARD] Reaction of Carbonyls with Semicarbazide — 9701/13/M/J/223/Q98**[1]**

What crystalline derivative is formed when an aldehyde reacts with semicarbazide ($\text{NH}_2\text{-NH-CONH}_2$)?

- [A] A semicarbazone, R-CH=N-NH-CONH_2 , and water
- [B] An ester
- [C] A carboxylic acid
- [D] A polyamide

99 [EASY] Summary of Carbonyl Identification Flowchart — 9701/11/M/J/224/Q99**[1]**

An unknown organic liquid X:

1. Forms an orange precipitate with 2,4-DNPH.
2. Gives NO mirror with Tollens' reagent.
3. Forms a yellow precipitate with alkaline aqueous iodine.
4. Has molecular formula $\text{C}_4\text{H}_8\text{O}$.

What is the identity of compound X?

- [A] Butanal
- [B] 2-methylpropanal
- [C] Butan-2-ol
- [D] Butanone, $\text{CH}_3\text{-CO-CH}_2\text{CH}_3$

100 [HARD] Iodoform Distinction of Phenethyl Alcohols — 9701/12/M/J/220/Q100**[1]**

Which compound will give a positive tri-iodomethane test: 1-phenylethanol ($\text{C}_6\text{H}_5\text{-CH(OH)-CH}_3$) or 2-phenylethanol ($\text{C}_6\text{H}_5\text{-CH}_2\text{-CH}_2\text{OH}$)?

- [A] 2-phenylethanol only.
- [B] 1-phenylethanol only, because it contains the required $\text{CH}_3\text{-CH(OH)-}$ methyl carbinol group.
- [C] Both compounds.
- [D] Neither compound.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 1: Deducing the Number of Chiral Centers in Cyanohydrin Synthesis — 9701/11/M/J/22/Q26**[1]**

Ethanal reacts with hydrogen cyanide, HCN, in the presence of a catalytic amount of NaCN to form 2-hydroxypropanenitrile.

Which statement concerning the product mixture is completely TRUE?

- [A] The product contains two chiral centers and is optically active.
- [B] The product is an optically active single enantiomer because cyanide attacks only from the top face.
- [C] The product contains one chiral carbon atom and is obtained as an optically inactive racemic mixture because the planar carbonyl group is attacked with equal probability from both faces.
- [D] The product contains no chiral centers and is achiral.

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Systematic Identification of Carbonyl Isomers with C₄H₈O — 9701/12/M/J/21/Q26**[1]**

There are three carbonyl isomers with the molecular formula C₄H₈O: butanal, 2-methylpropanal, and butanone.

Which single chemical reagent will produce a visible change with butanone, but will NOT produce a visible change with butanal or 2-methylpropanal?

- [A] Alkaline aqueous iodine (I₂ in aqueous NaOH)
- [B] Tollens' reagent
- [C] Fehling's solution
- [D] 2,4-dinitrophenylhydrazine

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Distinguishing Symmetrical from Unsymmetrical Ketones — 9701/13/O/N/22/Q26**[1]**

Consider two isomeric ketones with molecular formula C₅H₁₀O:

Compound P: Pentan-2-one; Compound Q: Pentan-3-one

Which chemical reagent produces an observable positive result with Compound P but shows NO reaction with Compound Q?

- [A] 2,4-dinitrophenylhydrazine
- [B] Acidified potassium dichromate(VI)
- [C] Tollens' reagent
- [D] Alkaline aqueous iodine (warm I₂ / NaOH)

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Stoichiometry of Hydride Reduction of Dicarboxylic Acids — 9701/11/F/M/22/Q26**[1]**

One mole of butane-2,3-dione, CH₃-CO-CO-CH₃, is completely reduced by sodium borohydride (NaBH₄) in aqueous ethanol.

How many moles of hydride ions (:H⁻) are transferred, and what is the classification of the resulting alcohol product?

- [A] 1 mole of hydride ions transferred; produces a primary diol
- [B] 2 moles of hydride ions transferred; produces butane-2,3-diol (a secondary diol)
- [C] 4 moles of hydride ions transferred; produces a primary alcohol
- [D] 2 moles of hydride ions transferred; produces a tertiary diol

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 5: The Diagnostic Reagents Flowchart for Unknown C₃ Compounds — 9701/12/O/N/21/Q26**[1]**

An unknown compound X of formula C₃H₆O is investigated:

- It forms an orange crystalline precipitate when shaken with 2,4-DNPH.
- When heated with Fehling's solution, the blue solution turns to a brick-red precipitate.

What is the identity of compound X, and what is the product when X is treated with NaBH₄?

- [A] X is propanal; reduction with NaBH₄ yields propan-1-ol.
- [B] X is propanone; reduction with NaBH₄ yields propan-2-ol.
- [C] X is prop-2-en-1-ol; reduction with NaBH₄ yields propan-1-ol.
- [D] X is methoxyethene; reduction with NaBH₄ yields ethanol.

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Chirality of Cyanohydrins Derived from Symmetrical vs Unsymmetrical Carbonyls — 9701/13/M/J/12/Q26

Two carbonyl compounds, ethanal and propanone, are each reacted separately with HCN in the presence of NaCN.

Which statement concerning the optical activity of the products is TRUE?

- [A] Both products exhibit optical isomerism.
- [B] Neither product can exhibit optical isomerism.
- [C] The product from ethanal can exhibit optical isomerism (exists as enantiomers), whereas the product from propanone cannot exhibit optical isomerism.
- [D] The product from propanone is optically active while that from ethanal is not.

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Quantitative Fehling's Solution Titration Analysis — 9701/11/O/N/21/Q26**[1]**

A sample of 0.020 mol of an unknown aliphatic aldehyde is heated with an excess of Fehling's solution. The resulting brick-red precipitate of copper(I) oxide (Cu_2O , $M_r = 143.0$) is collected, dried, and weighed.

What is the expected mass of Cu_2O precipitate? (Ar: Cu = 63.5, O = 16.0)

- [A] 1.43 g of Cu_2O
- [B] 2.86 g of Cu_2O
- [C] 5.72 g of Cu_2O
- [D] 0.72 g of Cu_2O

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Complete Oxidation and Hydrolysis Sequence of Carbonyl Derivatives — 9701/12/F/M/23/Q26**[1]**

Consider the reaction sequence:

Ethanal $\xrightarrow{\text{HCN / NaCN}}$ Compound P $\xrightarrow{\text{Dilute HCl, reflux}}$ Compound Q

What is the IUPAC name of Compound Q, and does it exhibit optical activity?

- [A] Propanoic acid; it is optically active.
- [B] 2-chloropropanoic acid; it is an optically active single enantiomer.
- [C] Aminoethanoic acid; it is achiral.
- [D] 2-hydroxypropanoic acid (lactic acid); it is obtained as a racemic mixture and is optically inactive.

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Predicting Products of Cross-Reaction of Carbonyls with Iodoform Reagent — 9701/13/M/J/23/Q26**[1]**

Which of the following compounds will NOT produce a yellow precipitate of tri-iodomethane when treated with I_2 and aqueous NaOH?

- [A] Methanal, HCHO
- [B] Ethanal, CH_3CHO
- [C] Propanone, CH_3COCH_3
- [D] Butanone, $\text{CH}_3\text{COCH}_2\text{CH}_3$

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Quantitative Percentage Composition Analysis of Carbonyls — 9701/11/M/J/23/Q28**[1]**

An organic carbonyl compound contains 66.7% carbon, 11.1% hydrogen, and 22.2% oxygen by mass.

- It forms an orange crystalline precipitate with 2,4-DNPH.
- It reduces Tollens' reagent to form a silver mirror.
- It does NOT give a yellow precipitate with alkaline iodine.

What is the systematic IUPAC name of this compound?

- [A] Butanone
- [B] Pentan-2-one
- [C] Butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ (or 2-methylpropanal)
- [D] Propanal

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. The carbonyl carbon forms three sigma bonds using sp ² hybrid orbitals in a planar trigonal arrangement (~120°). Sideways overlap of 2p orbitals forms the pi bond. Oxygen is much more electronegative than carbon (3.5 vs 2.5), pulling both sigma and pi electron density toward itself. This polarizes the bond (C delta+ = O delta-), making the carbonyl carbon vulnerable to nucleophilic attack.	[1]
2	D	Option D is correct. HCN has a tiny dissociation constant (K _a ~ 6 × 10 ⁻¹⁰ mol dm ⁻³) and dissociates negligibly in solution. The rate-determining first step of nucleophilic addition requires the attack of the nucleophile :CN ⁻ on the delta+ carbon. Adding a source of cyanide ions (NaCN or KCN) or adding OH ⁻ (OH ⁻ + HCN → H ₂ O + :CN ⁻) drastically boosts the concentration of :CN ⁻ , vastly increasing the reaction rate.	[1]
3	A	Option A is correct. The carbonyl carbon in propanal is sp ² hybridised and trigonal planar. The nucleophilic cyanide ion (:CN ⁻) can attack the planar C=O group with equal 50% probability from either the top face or bottom face. Attack from opposite faces generates mirror-image enantiomers of 2-hydroxybutanenitrile (CH ₃ CH ₂ -CH(OH)-CN) in exact 1:1 proportion, resulting in an optically inactive racemic mixture.	[1]
4	C	Option C is correct. Sodium borohydride (NaBH ₄) acts as a source of nucleophilic hydride ions (:H ⁻). Hydride addition to an aldehyde (butanal) reduces it to a primary alcohol (butan-1-ol). Hydride addition to a ketone (butanone) reduces it to a secondary alcohol (butan-2-ol).	[1]
5	A	Option A is correct. Carbonyl compounds have polar C=O bonds and exhibit permanent dipole-dipole attractions, giving higher boiling points than non-polar alkanes. However, because they lack hydrogen atoms bonded to oxygen, they cannot form intermolecular hydrogen bonds with each other, resulting in lower boiling points than alcohols.	[1]
6	D	Option D is correct. Although aldehydes and ketones cannot donate hydrogen bonds to each other, the lone pairs on the delta-oxygen readily accept hydrogen bonds from water molecules (C=O ... H-O-H). For small molecules, this interaction overcomes the energy needed to separate water molecules; for longer chains, the hydrophobic hydrocarbon tail prevents dissolution.	[1]
7	B	Option B is correct. Attack of :CN ⁻ pushes the pi electrons onto the carbonyl oxygen, forming a tetrahedral intermediate where the central carbon is sp ³ hybridised, bearing two methyl groups, a cyano group, and a negatively charged oxygen (-O ⁻).	[1]
8	C	Option C is correct. Acid-catalyzed hydrolysis of nitriles converts the -C#N group into a carboxylic acid (-COOH): CH ₃ -CH(OH)-CN + HCl + 2H ₂ O → CH ₃ -CH(OH)-COOH + NH ₄ Cl. This provides a key synthetic route to alpha-hydroxy acids.	[1]
9	B	Option B is correct. LiAlH ₄ reduces the nitrile group (-C#N) to a primary amine group (-CH ₂ NH ₂) while leaving the secondary alcohol group intact, forming the amino alcohol 1-aminopropan-2-ol.	[1]
10	A	Option A is correct. Ketones possess two electron-donating alkyl groups that disperse the delta+ charge on the carbonyl carbon via inductive effect (+I), reducing its electrophilicity. Furthermore, two bulky alkyl groups cause greater steric crowding around the carbonyl carbon in ketones compared to one alkyl group and a tiny hydrogen in aldehydes.	[1]
11	D	Option D is correct. The [BH ₄] ⁻ anion acts as a hydride donor: a hydride ion (:H ⁻ , a hydrogen with two electrons and a negative charge) attacks the partially positive carbonyl carbon, transferring a pair of electrons into a new C-H bond.	[1]
12	C	Option C is correct. LiAlH ₄ is a much more powerful reducing agent with a highly polarized Al-H bond. It reacts violently with protonic solvents (LiAlH ₄ + 4H ₂ O → LiOH + Al(OH) ₃ + 4H ₂), whereas NaBH ₄ reacts sluggishly with water/alcohols, allowing its safe use in aqueous ethanol.	[1]
13	A	Option A is correct. Aldehydes and unhindered methyl ketones react with concentrated NaHSO ₃ via nucleophilic addition of the bisulfite ion to form a crystalline, solid bisulfite adduct. Filtering the crystals and treating with acid or alkali regenerates the pure aldehyde, providing an effective purification technique.	[1]
14	C	Option C is correct. Methanal has two tiny hydrogen atoms attached to C=O (no +I inductive donation and zero steric crowding). The hydration equilibrium (H ₂ C=O + H ₂ O ⇌ H ₂ C(OH) ₂) lies >99.9% to the right. For propanone, two methyl groups stabilize the C=O group and steric crowding in the tetrahedral diol disfavors hydration (<0.1% hydrated).	[1]
15	B	Option B is correct. Methanal has two hydrogens on the carbonyl carbon. Addition of the carbanion R ⁻ from a Grignard reagent produces R-CH ₂ O ⁻ , which upon protonation yields a primary alcohol (R-CH ₂ OH) with one more carbon than the alkyl halide.	[1]
16	D	Option D is correct. Reaction with an aldehyde (R-CHO) introduces a second alkyl group to the carbonyl carbon, producing a secondary alcohol upon hydrolysis: R-CHO + R'MgX → R-CH(OMgX)-R' → R-CH(OH)-R'.	[1]
17	C	Option C is correct. A ketone already has two alkyl groups bonded to the carbonyl carbon. Addition of the third alkyl group from the Grignard reagent followed by protonation yields a tertiary alcohol: R ₂ C=O + R'MgX → R ₂ C(OH)R'.	[1]
18	A	Option A is correct. The carbonyl carbon forms three coplanar sigma bonds directed at ~120° (sp ²). The oxygen atom is also sp ² hybridised, utilizing one sp ² orbital to form the sigma bond with carbon and two sp ² hybrid orbitals to house its two non-bonding electron lone pairs in the molecular plane.	[1]
19	D	Option D is correct. The electronegativity difference between C (2.5) and O (3.5) creates a strong dipole (C delta+ = O delta-), making the carbon electrophilic and vulnerable to nucleophilic attack. In contrast, the symmetrical C=C bond has high electron density distributed above and below the axis, attracting electrophiles and repelling nucleophiles.	[1]
20	B	Option B is correct. Step 1 involves bringing two negatively/partially charged species together against electrostatic repulsion to form a high-energy transition state, overcoming the primary activation barrier. Step 2 (proton transfer from HCN to the alkoxide) is an extremely rapid acid-base neutralization.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
21	B	Option B is correct. Nucleophilic addition of an alcohol (R'OH) across the C=O bond of an aldehyde produces a hemiacetal, where the former carbonyl carbon is bonded to -H, -R, -OH, and -OR'.	[1]
22	D	Option D is correct. Reacting an aldehyde with excess alcohol and dry acid catalyst yields an acetal (R-CH(OR') ₂ + H ₂ O). Acetals are completely stable to strong bases, nucleophiles, and reducing agents (LiAlH ₄), serving to protect carbonyls during multi-step syntheses before being removed by dilute acid.	[1]
23	A	Option A is correct. Pure HCN is a lethal, fast-acting respiratory toxin with a low boiling point (26 °C). Using NaCN with controlled addition of dilute acid creates a buffered solution that minimizes toxic vapor release while providing high [CN ⁻] for fast reaction.	[1]
24	C	Option C is correct. The carbonyl group is conjugated with the aromatic ring. Resonance delocalises pi electrons from the benzene ring into the carbonyl carbon, dispersing its delta+ charge and diminishing its electrophilic reactivity towards nucleophiles.	[1]
25	A	Option A is correct. Each tert-butyl group consists of three methyl groups branching from a central carbon. Two such bulky groups create extreme steric congestion around the carbonyl carbon, creating an insurmountable steric barrier to nucleophilic attack.	[1]
26	D	Option D is correct. Ethanal boils at 20.8 °C. Conducting the oxidation in a distillation flask allows ethanal to vaporise instantly upon formation, escaping through the condenser into an ice-cooled receiver before it can be further oxidized to ethanoic acid.	[1]
27	B	Option B is correct. Unlike aldehydes, ketones lack a hydrogen atom attached to the carbonyl carbon and cannot be oxidized further without breaking C-C bonds. Refluxing ensures 100% conversion of the secondary alcohol into propanone without danger of over-oxidation.	[1]
28	C	Option C is correct. Volatile aldehydes are formed by incomplete fuel combustion and atmospheric photo-oxidation of hydrocarbons. They cause severe mucous membrane irritation and react with NO ₂ in sunlight to form PAN, a potent eye-watering component of photochemical smog.	[1]
29	B	Option B is correct. Steric crowding in the tetrahedral cyanohydrin product increases with alkyl substitution, and alkyl groups stabilize the starting carbonyl via +I inductive effect. Both factors shift the equilibrium increasingly to the left: HCHO (K very large) > CH ₃ CHO > CH ₃ COCH ₃ (K much smaller).	[1]
30	A	Option A is correct. The powerful electron-withdrawing inductive effect (-I) of six fluorine atoms pulls electron density away from the carbonyl carbon, creating an enormous delta+ charge. Water adds instantly to relieve this electron deficiency, and the gem-diol is thermodynamically favored.	[1]
31	D	Option D is correct. NaBH ₄ is a selective reducing agent that reduces polar carbonyl groups (>C=O) but does NOT reduce non-polar carbon-carbon double bonds (C=C). Thus, the aldehyde group is reduced to a primary alcohol while the alkene remains intact: CH ₃ -CH=CH-CH ₂ OH.	[1]
32	C	Option C is correct. Catalytic hydrogenation with H ₂ and Ni reduces both carbon-carbon double bonds (C=C → C-C) and carbonyl groups (C=O → CH-OH), saturating the entire molecule to yield butan-1-ol.	[1]
33	A	Option A is correct. Formaldehyde and ethanal have harsh, irritating, pungent odors. As chain length increases (e.g. octanal, nonanal, decanal, benzaldehyde), volatility decreases and molecular interactions with olfactory receptors yield pleasant citrus, rose, and almond aromas widely prized in perfumery.	[1]
34	C	Option C is correct. Hydroxylamine reacts with carbonyl compounds in an addition-elimination (condensation) reaction to form crystalline oximes: R-CHO + NH ₂ OH → R-CH=N-OH + H ₂ O.	[1]
35	B	Option B is correct. Hydrazine undergoes condensation with ketones to form hydrazones: R ₂ C=O + H ₂ N-NH ₂ → R ₂ C=N-NH ₂ + H ₂ O.	[1]
36	D	Option D is correct. The Wolff-Kishner reduction converts aldehydes and ketones into hydrocarbons: R ₂ C=O → R ₂ C=N-NH ₂ → [KOH, heat] → R ₂ CH ₂ + N ₂ (g).	[1]
37	C	Option C is correct. The Clemmensen reduction uses zinc amalgam (Zn(Hg)) and concentrated HCl to reduce carbonyl groups (>C=O) directly to methylene groups (-CH ₂ -): R ₂ C=O + 4[H] → R ₂ CH ₂ + H ₂ O.	[1]
38	A	Option A is correct. Tautomerism is a form of constitutional isomerism where isomers rapidly interconvert by shifting a proton from an alpha-carbon to the carbonyl oxygen: CH ₃ -C(=O)-CH ₃ ⇌ CH ₂ =C(OH)-CH ₃ . For simple ketones, the keto form overwhelmingly predominates (>99.9%).	[1]
39	D	Option D is correct. Loss of an alpha-proton produces an enolate anion [-CH ₂ -C(=O)-R ⇌ CH ₂ =C(O ⁻)-R]. The negative charge is shared between carbon and the highly electronegative oxygen atom, providing massive resonance stabilization that lowers pK _a by 30 units compared to alkanes.	[1]
40	B	Option B is correct. Dilute base deprotonates one ethanal molecule to form an enolate ion, which attacks the carbonyl carbon of a second ethanal molecule: CH ₃ CHO + CH ₃ CHO → CH ₃ -CH(OH)-CH ₂ -CHO (3-hydroxybutanal, an aldol containing alcohol and aldehyde groups).	[1]
41	B	Option B is correct. Propanone is symmetrical ((CH ₃) ₂ C=O). Attack by cyanide yields (CH ₃) ₂ C(OH)CN. Carbon-2 is bonded to -OH, -CN, and TWO identical -CH ₃ groups. Because it lacks four different groups, the molecule is achiral and cannot show optical activity under any circumstances.	[1]
42	D	Option D is correct. 3-methylbutan-2-ol is a secondary alcohol: (CH ₃) ₂ CH-CH(OH)-CH ₃ . Reduction of a secondary alcohol must originate from the corresponding ketone: (CH ₃) ₂ CH-CO-CH ₃ (3-methylbutan-2-one).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
43	A	Option A is correct. The product is a primary alcohol ($-\text{CH}_2\text{OH}$): $(\text{CH}_3)_3\text{C}-\text{CH}_2\text{OH}$. Primary alcohols are formed exclusively by the reduction of aldehydes. Therefore, W is the aldehyde 2,2-dimethylpropanal $((\text{CH}_3)_3\text{C}-\text{CHO})$.	[1]
44	C	Option C is correct. Both undergo initial nucleophilic attack to form a tetrahedral intermediate. In acyl chlorides, the chloride ion is a stable weak base and departs readily (addition-elimination = net substitution). In aldehydes, H^- is an intensely strong base and cannot leave, forcing the alkoxide to be protonated (net addition).	[1]
45	A	Option A is correct. Nitrile hydrolysis requires 2 molecules of water in the presence of acid: the carbon of the $-\text{CN}$ group becomes a carboxylic acid ($-\text{COOH}$) and nitrogen is released as the ammonium ion (NH_4^+): $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$.	[1]
46	D	Option D is correct. Benzaldehyde undergoes nucleophilic addition with $:\text{CN}^-$ across its carbonyl group to yield 2-hydroxy-2-phenylethanenitrile (mandelonitrile), which can be hydrolyzed to mandelic acid.	[1]
47	B	Option B is correct. The mechanism requires both $:\text{CN}^-$ (nucleophile for Step 1) and HCN (proton donor for Step 2). In strong acid, CN^- is protonated to HCN (no nucleophile). In strong base, all HCN is deprotonated to CN^- (no proton donor for Step 2). Optimal rate occurs at pH 8–9.	[1]
48	C	Option C is correct. Protonating oxygen generates the resonance contributor $>\text{C}^+-\text{OH}$, placing a full formal positive charge on carbon, dramatically accelerating attack by weak neutral nucleophiles (such as water or alcohols).	[1]
49	B	Option B is correct. The $-\text{I}$ inductive effect of three chlorine atoms severely destabilizes the carbonyl group by increasing δ^+ on carbon, while the two $-\text{OH}$ groups in the gem-diol form stabilizing intramolecular hydrogen bonds with chlorine atoms.	[1]
50	A	Option A is correct. Heated copper acts as a dehydrogenation catalyst: $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CHO} + \text{H}_2(\text{g})$. In the absence of oxygen, no further oxidation to carboxylic acid can occur, giving pure aldehyde.	[1]
51	D	Option D is correct. In concentrated aqueous solution, methanal undergoes reversible addition polymerisation to form linear polyoxymethylene polymers known as paraformaldehyde, which decompose back to gaseous formaldehyde upon gentle heating.	[1]
52	C	Option C is correct. The starting material already possesses a chiral carbon at C_2 (bonded to $-\text{H}$, $-\text{Cl}$, $-\text{CH}_3$, $-\text{CHO}$). Nucleophilic addition of HCN creates a new chiral center at the former carbonyl carbon ($-\text{CH}(\text{OH})-\text{CN}$), yielding a diastereomeric product with 2 chiral stereocenters.	[1]
53	A	Option A is correct. LiAlH_4 reduces esters by cleaving the ester linkage and reducing the acyl fragment: $\text{CH}_3\text{COOCH}_2\text{CH}_3 + 4[\text{H}] \rightarrow 2 \text{CH}_3\text{CH}_2\text{OH}$. Both fragments yield ethanol.	[1]
54	C	Option C is correct. A double bond has four shared electrons and an exposed, highly polarizable pi cloud. The dipole moment of $\text{C}=\text{O}$ is significantly amplified by pi polarization toward oxygen compared to the single sigma $\text{C}-\text{Cl}$ bond.	[1]
55	B	Option B is correct. Quantum orbital overlap requires the nucleophile's lone pair to overlap with the empty antibonding pi^* orbital of the carbonyl group, which is oriented at $\sim 107^\circ$ relative to the $\text{C}=\text{O}$ double bond axis.	[1]
56	D	Option D is correct. 2,4-dinitrophenylhydrazine (2,4-DNPH / Brady's reagent) undergoes a condensation reaction specifically with the carbonyl group ($>\text{C}=\text{O}$) of aldehydes and ketones, eliminating water to precipitate a brightly colored (yellow to red-orange) crystalline 2,4-dinitrophenylhydrazone derivative.	[1]
57	C	Option C is correct. Because many aldehydes and ketones have very similar boiling points, direct identification can be difficult. The crystalline 2,4-dinitrophenylhydrazone derivatives have sharp, distinct melting points. Purifying by recrystallisation, drying, and measuring the melting point allows unambiguous identification by looking up reference tables.	[1]
58	A	Option A is correct. Tollens' reagent is an alkaline solution of diamminesilver(I) complex ions, $[\text{Ag}(\text{NH}_3)_2]^+$. Aldehydes are easily oxidized to carboxylate salts and reduce the silver ions to metallic silver ($\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s})$), coating the clean glass test tube with a brilliant silver mirror. Ketones cannot be oxidized mildly and give no reaction.	[1]
59	D	Option D is correct. Fehling's solution contains Cu^{2+} complexed with tartrate ions in alkaline medium (deep blue). Aliphatic aldehydes reduce $\text{Cu}(\text{II})$ to $\text{Cu}(\text{I})$, precipitating brick-red copper(I) oxide (Cu_2O): $\text{RCHO} + 2\text{Cu}^{2+} + 5\text{OH}^- \rightarrow \text{RCOO}^- + \text{Cu}_2\text{O}(\text{s}) + 3\text{H}_2\text{O}$. Ketones cannot reduce Fehling's solution and the solution remains blue.	[1]
60	B	Option B is correct. The tri-iodomethane reaction tests specifically for the methyl carbonyl group, $\text{CH}_3-\text{C}(=\text{O})-$ (or secondary methyl carbinols that oxidize to it). In aldehydes, ONLY ethanal (CH_3CHO) has this group. In ketones, all methyl ketones (e.g. propanone, butanone, pentan-2-one) possess this group and form a yellow precipitate of CHI_3 .	[1]
61	B	Option B is correct. An aldehyde has general formula $\text{R}-\text{CHO}$. For it to possess the $\text{CH}_3-\text{C}(=\text{O})-$ group, R must be a methyl group ($-\text{CH}_3$). Thus, CH_3CHO (ethanal) is the sole aldehyde meeting this structural criterion; all other aldehydes have $\text{R} = \text{H}$ (methanal) or longer alkyl chains lacking a methyl carbonyl.	[1]
62	D	Option D is correct. Pentan-2-one has the methyl carbonyl structure $\text{CH}_3-\text{CO}-$ and gives a positive iodoform test (yellow CHI_3 precipitate). Pentan-3-one and hexan-3-one have ethyl or propyl groups on both sides of the carbonyl (no methyl group attached to $\text{C}=\text{O}$), giving a negative test.	[1]
63	A	Option A is correct. Reaction: $\text{CH}_3\text{COCH}_3 + 3\text{I}_2 + 4\text{NaOH} \rightarrow \text{CHI}_3(\text{s}) + \text{CH}_3\text{COONa} + 3\text{NaI} + 3\text{H}_2\text{O}$. Cleavage of the tri-iodinated intermediate $\text{CI}_3-\text{CO}-\text{CH}_3$ by hydroxide yields yellow solid tri-iodomethane (CHI_3) and the salt sodium ethanoate (CH_3COONa).	[1]
64	C	Option C is correct. The aldehyde is oxidized to a carboxylate anion, releasing two electrons: $\text{RCHO} + 3\text{OH}^- \rightarrow \text{RCOO}^- + 2\text{H}_2\text{O} + 2\text{e}^-$. The silver diamine complex ion is reduced by accepting an electron to deposit metallic silver: $[\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^- \rightarrow \text{Ag}(\text{s}) + 2\text{NH}_3$.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
65	A	Option A is correct. The resonance delocalization from the benzene ring stabilizes benzaldehyde against oxidation. Tollens' reagent is a stronger oxidizing agent (E° higher) and oxidizes benzaldehyde to benzoate. Fehling's solution is milder and only oxidizes aliphatic aldehydes.	[1]
66	D	Option D is correct. Aldehydes are readily oxidized to carboxylic acids: $3\text{CH}_3\text{CHO} + \text{Cr}_2\text{O}_7^{2-} + 8\text{H}^+ \rightarrow 3\text{CH}_3\text{COOH} + 2\text{Cr}^{3+} + 4\text{H}_2\text{O}$. The orange dichromate(VI) ions are reduced to green chromium(III) ions.	[1]
67	B	Option B is correct. Ketones lack a hydrogen atom bonded to the carbonyl carbon. Under standard mild acidic conditions, they cannot be oxidized without breaking strong C-C bonds, so the orange dichromate solution remains unchanged.	[1]
68	C	Option C is correct. 2,4-DNPH reacts with ALL carbonyl compounds (both aldehydes and ketones), giving an orange precipitate with both propanal and propanone. Hence, 2,4-DNPH cannot distinguish between them; Tollens', Fehling's, or acidified $\text{K}_2\text{Cr}_2\text{O}_7$ must be used.	[1]
69	B	Option B is correct. Both ethanal and propanal are aldehydes: both form silver mirrors with Tollens', brick-red precipitates with Fehling's, and orange precipitates with 2,4-DNPH. However, ethanal is a methyl carbonyl (CH_3CHO) and gives a yellow precipitate with alkaline iodine, whereas propanal ($\text{CH}_3\text{CH}_2\text{CHO}$) lacks a methyl carbonyl and gives no precipitate.	[1]
70	A	Option A is correct. Both are ketones (no reaction with Tollens' or Fehling's; both react with 2,4-DNPH). Propanone (CH_3COCH_3) is a methyl ketone and gives a positive iodoform test; pentan-3-one ($\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$) is not a methyl ketone and gives a negative test.	[1]
71	D	Option D is correct. The lone pair on the terminal hydrazine nitrogen attacks the carbonyl carbon (nucleophilic addition). Subsequent proton transfer and elimination of a water molecule (H_2O) creates a C=N double bond: $>\text{C}=\text{O} + \text{H}_2\text{N}-\text{NH}-\text{Ar} \rightarrow >\text{C}=\text{N}-\text{NH}-\text{Ar} + \text{H}_2\text{O}$ (an addition-elimination condensation).	[1]
72	C	Option C is correct. 2,4-DNPH consists of a benzene ring bearing a hydrazine group ($-\text{NH}-\text{NH}_2$) at position 1, with electron-withdrawing nitro groups ($-\text{NO}_2$) at positions 2 and 4 (ortho and para to the hydrazine group).	[1]
73	A	Option A is correct. Derivatives of simple hydrazine are often liquids or low-melting oils that are difficult to isolate. The two nitro groups and aromatic ring in 2,4-DNPH create high-melting, brightly colored crystalline solids that crystallize readily and have distinct, tabulated melting points.	[1]
74	C	Option C is correct. Recrystallisation involves dissolving the crude solid in the minimum volume of boiling solvent, hot-filtering any insoluble grit, allowing the filtrate to cool slowly so pure crystals form while soluble impurities stay in the mother liquor, filtering under vacuum (Buchner funnel), and washing with ice-cold solvent.	[1]
75	B	Option B is correct. A tiny plug of dry powder is sealed into a capillary tube and heated in an electrical melting point apparatus. Heating must be slow ($\sim 1^\circ\text{C}/\text{min}$) near the expected transition to record the exact melting temperature accurately.	[1]
76	D	Option D is correct. Procedure: $2\text{AgNO}_3 + 2\text{NaOH} \rightarrow \text{Ag}_2\text{O}(\text{s}) + 2\text{NaNO}_3 + \text{H}_2\text{O}$ (brown precipitate of silver(I) oxide). Dilute aqueous ammonia is added dropwise with shaking: $\text{Ag}_2\text{O}(\text{s}) + 4\text{NH}_3 + \text{H}_2\text{O} \rightarrow 2[\text{Ag}(\text{NH}_3)_2]^+ + 2\text{OH}^-$. Ammonia is added until the precipitate just dissolves to give a clear, colorless solution of $[\text{Ag}(\text{NH}_3)_2]^+$.	[1]
77	C	Option C is correct. On prolonged standing or drying, ammoniacal silver solutions form silver nitride (Ag_3N) and silver imide (Ag_2NH), known as 'fulminating silver'. These dark precipitates are violently contact-explosive even when touched with a feather, so Tollens' reagent must always be prepared fresh and safely disposed of immediately.	[1]
78	A	Option A is correct. A mirror requires silver crystals to nucleate smoothly onto clean glass surface silanol groups. If grease or oil is present, silver cannot adhere uniformly and precipitates into solution as a dull black/grey colloidal suspension.	[1]
79	D	Option D is correct. Fehling's test reduces blue Cu^{2+} (+2 state) to copper(I) oxide (Cu_2O , where copper is in the +1 state). Cu_2O is an insoluble brick-red / orange-red crystalline solid.	[1]
80	B	Option B is correct. Fehling's solution must be stored as two separate solutions (Fehling's A and B) and mixed freshly because alkaline tartrate degrades over time. Benedict's solution uses sodium citrate and carbonate; it is less alkaline and stable indefinitely as a single pre-mixed reagent, widely used in clinical urine testing for glucose.	[1]
81	B	Option B is correct. Methanal is unique: it first oxidizes to methanoic acid (HCOOH , reducing $2\text{Cu}^{2+} \rightarrow 1\text{Cu}_2\text{O}$). Methanoic acid still possesses an oxidizable aldehyde-like C-H bond, which is further oxidized to CO_2 (reducing another $2\text{Cu}^{2+} \rightarrow 1\text{Cu}_2\text{O}$). Thus, 1 mole of HCHO reduces 4 moles of Cu^{2+} , producing 2 moles of Cu_2O .	[1]
82	D	Option D is correct. Butan-2-ol is a secondary alcohol: it is oxidized by acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (orange $\rightarrow$ green) and reacts with sodium to evolve H_2 , but does NOT react with 2,4-DNPH. Butanone is a ketone: it resists $\text{K}_2\text{Cr}_2\text{O}_7$ (stays orange), does not react with Na, but produces an immediate orange precipitate with 2,4-DNPH.	[1]
83	A	Option A is correct. Tri-iodomethane has formula CHI_3 : 1 Carbon (12.0) + 1 Hydrogen (1.0) + 3 Iodine ($3 \times 126.9 = 380.7$) = 393.7 g/mol. Its high molecular mass accounts for its solid state and low volatility.	[1]
84	C	Option C is correct. Reacting with 2,4-DNPH confirms a carbonyl ($>\text{C}=\text{O}$). No reaction with Tollens' proves it is a ketone (not an aldehyde). A yellow precipitate with alkaline iodine proves it is a methyl ketone ($\text{CH}_3-\text{CO}-$). The 5-carbon methyl ketones are pentan-2-one ($\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$) and 3-methylbutan-2-one ($\text{CH}_3\text{COCH}(\text{CH}_3)_2$).	[1]
85	A	Option A is correct. In carboxylic acids and esters, resonance delocalization of the lone pair on the adjacent oxygen into the $\text{C}=\text{O}$ group stabilizes the carbon and dramatically diminishes its electrophilicity. Hence, only unhindered aldehydes and ketones react with 2,4-DNPH.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
86	D	Option D is correct. In aromatic/conjugated hydrazones, the pi system extends continuously across the entire molecule. Increased pi delocalization decreases the energy required for electronic pi → pi* transitions, absorbing blue/green light and transmitting red/orange light.	[1]
87	B	Option B is correct. Gentle warming ensures smooth halogenation. Boiling causes loss of volatile reactants (ethanal, propanone) and causes the active oxidizing species (hypoiodite, IO ⁻) to disproportionate rapidly into iodide and iodate: 3IO ⁻ → 2I ⁻ + IO ₃ ⁻ , quenching the test.	[1]
88	C	Option C is correct. Phenylethanone contains a methyl carbonyl group attached to a phenyl ring (C ₆ H ₅ -CO-CH ₃). It undergoes tri-iodination and alkaline cleavage to precipitate yellow CHI ₃ , leaving sodium benzoate (C ₆ H ₅ COONa) in solution.	[1]
89	B	Option B is correct. Silver is unreactive to non-oxidizing acids but dissolves cleanly in concentrated nitric acid to regenerate silver nitrate solution (AgNO ₃), which can be neutralized and reused.	[1]
90	A	Option A is correct. Propan-1-ol, propan-2-ol, and propanal are all oxidizable by K ₂ Cr ₂ O ₇ . However, propanal is the only aldehyde: warming with Tollens' reagent immediately produces a silver mirror, while propan-1-ol and propan-2-ol produce no reaction.	[1]
91	D	Option D is correct. Propan-2-ol contains the CH ₃ -CH(OH)- methyl carbinol group and gives a positive iodoform test (yellow precipitate of CHI ₃). Propan-1-ol lacks this group and gives no precipitate.	[1]
92	C	Option C is correct. The stoichiometry is 1 : 1 between a mono-methyl ketone and tri-iodomethane: 0.010 mol of methyl ketone yields 0.010 mol of CHI ₃ . Mass = 0.010 mol * 393.7 g/mol = 3.937 g ~ 3.94 g.	[1]
93	A	Option A is correct. In alkaline conditions (pH > 11), Cu ²⁺ ions normally precipitate instantly as insoluble gelatinous copper(II) hydroxide, Cu(OH) ₂ . Tartrate ions form a soluble deep-blue bistartratecuprate(II) complex, keeping Cu ²⁺ in solution so it is available to react with aldehydes upon heating.	[1]
94	C	Option C is correct. In the presence of hydroxide ions, silver ions precipitate as brown Ag ₂ O. Adding ammonia converts Ag ⁺ into the soluble coordinate complex [Ag(NH ₃) ₂] ⁺ , maintaining a controlled, low concentration of free Ag ⁺ ions in solution.	[1]
95	B	Option B is correct. Mr = 72.0 corresponds to C ₄ H ₈ O (4*12 + 8*1 + 16 = 72). Reaction with Fehling's proves it is an aldehyde (ruling out butanone). Negative iodoform test proves it is NOT ethanal (which has Mr = 44 anyway). Both butanal and 2-methylpropanal are aldehydes of Mr = 72 that give brick-red precipitates with Fehling's and negative iodoform tests.	[1]
96	D	Option D is correct. Both butanal and 2-methylpropanal are aliphatic aldehydes with formula C ₄ H ₈ O: both react identically with Fehling's, Tollens', K ₂ Cr ₂ O ₇ , and alkaline iodine. They can only be distinguished by the precise melting points of their crystalline 2,4-DNPH derivatives (or by NMR/mass spec fragmentation).	[1]
97	C	Option C is correct. Schiff's reagent is fuchsine dye decolorized by sulfurous acid. Aliphatic aldehydes react rapidly at room temperature to restore the intense magenta-purple coloration, serving as an exceptionally sensitive test for aldehydes (ketones do not react or react very sluggishly).	[1]
98	A	Option A is correct. Semicarbazide undergoes condensation with carbonyl compounds to eliminate water, forming crystalline semicarbazones: R-CHO + H ₂ N-NH-CONH ₂ → R-CH=N-NH-CONH ₂ + H ₂ O, which have sharp melting points.	[1]
99	D	Option D is correct. Step 1 confirms a carbonyl (>C=O). Step 2 confirms a ketone (not an aldehyde). Step 3 confirms a methyl ketone (CH ₃ -CO-). Formula C ₄ H ₈ O with a methyl ketone group must be butanone (CH ₃ -CO-CH ₂ CH ₃).	[1]
100	B	Option B is correct. 1-phenylethanol has a methyl group directly bonded to the carbinol carbon (C ₆ H ₅ -CH(OH)-CH ₃), which is oxidized by alkaline iodine to the methyl ketone acetophenone and forms yellow CHI ₃ . 2-phenylethanol is a primary alcohol lacking the methyl carbinol unit and gives a negative test.	[1]
101	C	Option C is correct. The addition of :CN ⁻ to ethanal yields 2-hydroxypropanenitrile: CH ₃ -CH(OH)-CN. The central carbon is bonded to four different groups (-H, -OH, -CH ₃ , -CN), making it a chiral stereocenter. Because the starting carbonyl carbon is sp ² hybridised and planar, nucleophilic attack by :CN ⁻ occurs with identical 50% probability from either face, yielding an equimolar racemic mixture that shows zero net optical rotation.	[1]
102	A	Option A is correct. Butanal and 2-methylpropanal are aldehydes, whereas butanone is a ketone. All three react with 2,4-DNPH. Both aldehydes react with Tollens' (silver mirror) and Fehling's (brick-red precipitate), while butanone does not. However, butanone contains a methyl carbonyl group (CH ₃ -CO-CH ₂ CH ₃) and reacts with alkaline iodine to precipitate yellow tri-iodomethane (CHI ₃). Butanal and 2-methylpropanal lack the methyl carbonyl group and produce no precipitate with alkaline iodine.	[1]
103	D	Option D is correct. Both pentan-2-one and pentan-3-one are ketones: neither reacts with Tollens' reagent or acidified K ₂ Cr ₂ O ₇ , and both form orange precipitates with 2,4-DNPH. However, pentan-2-one (CH ₃ COCH ₂ CH ₂ CH ₃) has a methyl carbonyl group and forms a yellow precipitate of tri-iodomethane with alkaline iodine. Pentan-3-one (CH ₃ CH ₂ COCH ₂ CH ₃) is a symmetrical diethyl ketone lacking a methyl carbonyl group, giving a negative test.	[1]
104	B	Option B is correct. Butane-2,3-dione contains two ketone carbonyl groups. Each ketone group requires one hydride ion (:H ⁻) to reduce it to a secondary alcohol: CH ₃ -CO-CO-CH ₃ + 2[H ⁻] → CH ₃ -CH(OH)-CH(OH)-CH ₃ . Thus, 2 moles of hydride ions are consumed, producing the secondary diol butane-2,3-diol.	[1]
105	A	Option A is correct. An orange precipitate with 2,4-DNPH proves X is a carbonyl compound (aldehyde or ketone). A brick-red precipitate with Fehling's solution uniquely proves X is an aldehyde (propanal, CH ₃ CH ₂ CHO). Reduction of the aldehyde propanal with NaBH ₄ produces the primary alcohol propan-1-ol (CH ₃ CH ₂ CH ₂ OH).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
106	C	Option C is correct. Ethanal yields 2-hydroxypropanenitrile, $\text{CH}_3\text{-CH(OH)-CN}$. The central carbon has four different groups ($-\text{H}$, $-\text{OH}$, $-\text{CH}_3$, $-\text{CN}$) and is chiral, allowing optical isomerism. Propanone yields 2-hydroxy-2-methylpropanenitrile, $(\text{CH}_3)_2\text{C(OH)CN}$. The central carbon is bonded to TWO identical methyl groups ($-\text{CH}_3$), so it is achiral and cannot show optical isomerism.	[1]
107	B	Option B is correct. Reaction equation: $\text{RCHO} + 2\text{Cu}^{2+} + 5\text{OH}^- \rightarrow \text{RCOO}^- + \text{Cu}_2\text{O(s)} + 3\text{H}_2\text{O}$. Exactly 1 mole of Cu_2O is produced per mole of mono-aldehyde. Thus, 0.020 mol of aldehyde produces 0.020 mol of Cu_2O . Mass = $0.020 \text{ mol} \times 143.0 \text{ g/mol} = 2.86 \text{ g}$.	[1]
108	D	Option D is correct. Ethanal reacts with HCN to yield 2-hydroxypropanenitrile (P, $\text{CH}_3\text{CH(OH)CN}$) as a racemic mixture. Refluxing with dilute HCl hydrolyzes the nitrile group to a carboxylic acid, forming 2-hydroxypropanoic acid (Q, lactic acid, $\text{CH}_3\text{CH(OH)COOH}$). Because the starting cyanohydrin was formed as a 50:50 racemic mixture, compound Q is also obtained as an equimolar racemic mixture and is optically inactive.	[1]
109	A	Option A is correct. The tri-iodomethane reaction requires the methyl carbonyl group, $\text{CH}_3\text{-C(=O)-}$. Methanal has formula H-C(=O)-H and has no methyl group attached to the carbonyl carbon, producing zero precipitate. Ethanal (CH_3CHO), propanone (CH_3COCH_3), and butanone ($\text{CH}_3\text{COCH}_2\text{CH}_3$) all contain the $\text{CH}_3\text{-C(=O)-}$ unit and form yellow precipitates of CHI_3 .	[1]
110	C	Option C is correct. Moles of C = $66.7/12.0 = 5.56$; H = $11.1/1.0 = 11.1$; O = $22.2/16.0 = 1.3875$. Dividing by 1.3875: C: $5.56/1.3875 = 4.0$; H: $11.1/1.3875 = 8.0$; O: $1.3875/1.3875 = 1.0$. Empirical formula is $\text{C}_4\text{H}_8\text{O}$ ($M_r = 72.0$). Orange precipitate with 2,4-DNPH confirms a carbonyl; silver mirror with Tollens' confirms an aldehyde (ruling out butanone); negative iodoform test is consistent with an unbranched or branched 4-carbon aldehyde (butanal or 2-methylpropanal).	[1]